Track 1 report — one child, one gene, two broken copies

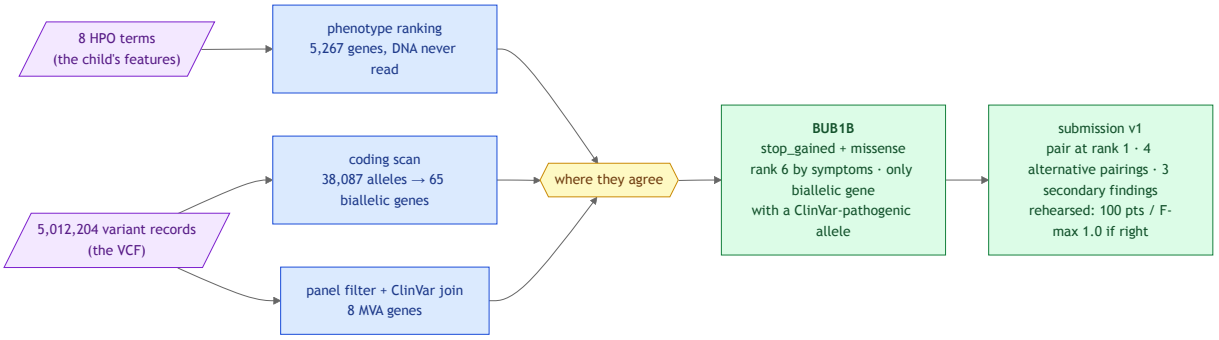
Standing: the Track-1 submission report, written 2026-09-05 from executed runs. Every number below is owned by `ideation/data-findings.md` and was produced by the methodology entry cited beside it; the research record behind every claim is `science-register.json`, rendered in `figures/science-report.html`. **Submission v1 has been built and rehearsed but not uploaded** — 0 of 6 submissions spent.

Written for a reader who knows software, not genetics. Each domain term is given as `technical term (plain meaning)` on first use; the full vocabulary is `dictionary.md`.

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1. The answer in one paragraph

The child has **mosaic variegated aneuploidy (MVA)** — a condition in which cells randomly gain or lose whole chromosomes because the machinery that checks chromosome separation is weak. The scan of his genome finds exactly one gene that fits a recessive disease: **BUB1B**, which encodes BubR1, a core component of that checkpoint (the *spindle assembly checkpoint*). He carries two different damaged copies of it, one inherited from each parent (a **compound heterozygote**): a **stop-gain variant** (a "premature full stop" that truncates the protein; already classified pathogenic for MVA syndrome 1 in the public ClinVar database) and a **missense variant** (one amino acid swapped) in the protein's kinase domain that no population database has ever seen and that AlphaMissense calls likely pathogenic. Four independent instruments converge on this pair — a gene-panel filter, a public-database join, a genome-wide scan of every protein-coding gene, and a symptom-only ranking that never read the DNA — and the raw-read, filtered-call and structural-variant alternatives were checked and found empty within their scope. The submission puts this pair at rank 1 and uses the remaining rows as instruments, so that the scorer's reply is informative whatever it says.



> **Legend** — **purple = raw input** (what the challenge delivered); **blue = machinery** (each an independent instrument); **yellow = the decision point** (convergence); **green = the goal** (the gene, the pair, the file).

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2. The clinical picture, and what it predicts

The challenge supplied eight **HPO terms** (Human Phenotype Ontology — a controlled vocabulary of clinical features, each with a stable identifier) and nothing else: no age, sex, karyotype or parental DNA (`ressources/hackathon-rules/`).

Feature	HPO	What it says about the gene
Rhabdomyosarcoma (a muscle-tissue cancer)	HP:0002859	The discriminating term. Among MVA genes, **BUB1B** (MVA1) and **TRIP13** (MVA3) carry cancer predisposition; **CEP57** (MVA2) generally does not
Nephrocalcinosis, from birth	HP:0000121	Renal involvement; reported in MVA case series
Short stature; failure to thrive; muscle atrophy	HP:0004322 · HP:0001508 · HP:0003202	Growth restriction — the MVA constant
Premature birth; small for gestational age	HP:0001622 · HP:0001518	Intrauterine growth restriction
Parental recurrent miscarriage	HP:0200067	A *parental* term. It argues for **biallelic** inheritance (both parents carriers of a chromosome-instability allele) — and it is not a feature of the child, so it must not enter the child's ranking (H002)

MVA is caused by biallelic (both copies) variants in a handful of spindle-assembly-checkpoint genes; the prior before any DNA is read is *compound heterozygous in BUB1B*, fallback TRIP13 (intel §3). Two things about the data sharpen it:

- **Sex inferred male** from the callset — hemizygous X, Y present at real depth (data-findings §5).
- **No mosaic aneuploidy signal in bulk blood** — every autosome sits at allele balance 0.5 and ~44× depth (037). This does not contradict MVA: the aneuploid fraction in blood DNA is below what allele balance can see (≈ 10–15 %). Reported as a bounded negative, not hidden.

A symptom-only ranker built for this project (Exomiser's human-phenotype path re-implemented in Python, reading two public ontology files and never the callset) places **BUB1B 6th of 5,267 curated genes** on the child's seven own terms, and reproduces the panel's own confidence order without being shown it — DICER1 8, TP53 11, CEP57 17, TRIP13 26 (067). Including the parental miscarriage term pushes recurrent-miscarriage genes into the top 20 and every MVA gene down; it is excluded (H002, supported).

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3. The data, and what it can and cannot show

Delivered	What it is	Consequence
`WGS_EX2312012_HGWCNDSX7.vcf.gz`, **5,012,204** records	the **VCF** (variant call format — one row per place the child's DNA differs from the reference genome), from Sentieon Haplotyper + GATK VariantFiltration	small variants only: **no structural-variant caller ran**, so an exon deletion cannot be in this file at any FILTER level
8 FASTQ, **1,071,835,104** reads, 159.7 Gb	the raw sequencer output	the only route to an allele the caller never emitted; not needed for the answer below
no BAM	—	any read-level work means aligning ourselves

Three facts that shaped the pipeline (data-findings §2): contigs are bare (`15`, not `chr15`) while the scorer wants `chr15` and compares strings verbatim; the sample is not called `PROBAND01`; and the VCF carries **no annotation** — its `INFO/AF` is within-sample (0.5 or 1.0), not a population frequency. Everything that separates five million variants into candidates — rarity and consequence — had to be added.

Callset quality is normal for a human genome: 4.09 M SNVs, 0.93 M indels, ts/tv 1.94, het/hom 1.61, 66,860 multi-allelic sites (data-findings §3).

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4. The platform (what was built before the question was asked)

Every byte the challenge delivered is hash-verified in an object store; the VCF is loaded losslessly as text into PostgreSQL (`warehouse.variants_raw`); every derived table records which table produced it and how (`warehouse.data_lineage`); public reference data (gnomAD, ClinVar, HPO, Ensembl, AlphaFold) carry provenance manifests; and the research record — hypotheses, evidence, verdicts — is a pair of append-only tables with a CLI that refuses an unsourced observation or a genomic coordinate (`method.md`, `toolset.md`). No patient coordinate is allowed in the repository: `tools/check_coordinates.py` scans every tracked file for the candidate positions and the candidate gene's span and runs inside the quality gate; the one earlier entry that had quoted the pair was redacted when the gate first ran (075). The submission file lives outside the repository.

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5. Step by step — how five million became two

5.1 The panel, annotated (step 2, entries 031 and 051)

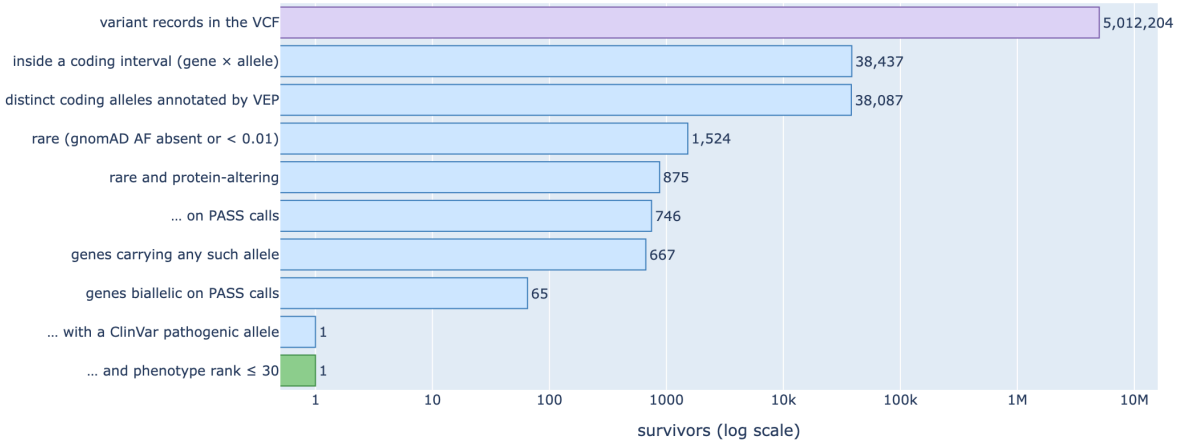
The eight MVA/differential genes hold 861 records (874 alleles after splitting multi-allelic sites). Ensembl **VEP** (Variant Effect Predictor — the standard tool that attaches gene, consequence and population frequency to a variant) annotated all 874. Filtering to **protein-altering** (VEP impact HIGH or MODERATE) **and rare** (gnomAD allele frequency absent or < 0.01, the recessive threshold) leaves **exactly two alleles, both in BUB1B, both heterozygous, both PASS** (data-findings §6.1). Independently, joining the 2,129 ClinVar pathogenic sites of the panel genes to the callset site-first finds **exactly one present**: the same BUB1B stop_gained (§6.2). Registered as **H003 (retrospective), supported 0.93**.

5.2 Genome-wide, without a panel (step 2, entries 069–070) — new this week

A panel is a list someone chose. The honest test is the whole genome. Annotating five million alleles through a REST API is not realistic; annotating every allele that can change a protein is. So:

1. **Coding intervals.** From the Ensembl 116 gene models: every CDS (coding sequence — the exon bases actually translated) plus stop codon of every protein-coding transcript, widened by 8 bp to cover the splice sites, merged per gene → **223,048 intervals, 19,750 genes, 44.6 Mb** (1.4 % of the genome). Case-agnostic tool with a provenance manifest: `tools/build_coding_regions.py`.
2. **Layer 2.** A binned range join of the callset to those intervals → `results.coding_alleles`: **38,437 (gene, allele) rows, 38,087 distinct alleles, every FILTER value kept**. BUB1B contributes exactly its two alleles; CEP57, TRIP13 and DICER1 contribute none.
3. **Annotation.** `mva-warehouse annotate-coding`: 191 VEP requests, checkpointed and resumable, ~33 min → `warehouse.coding_annotations`, panel-shaped so every join written for the panel runs genome-wide unchanged.

4. **Gene table.** `results.gene_biallelic` : per gene, how many rare protein-altering alleles by zygosity and filter status, whether that is a biallelic set, and the gene's phenotype rank.

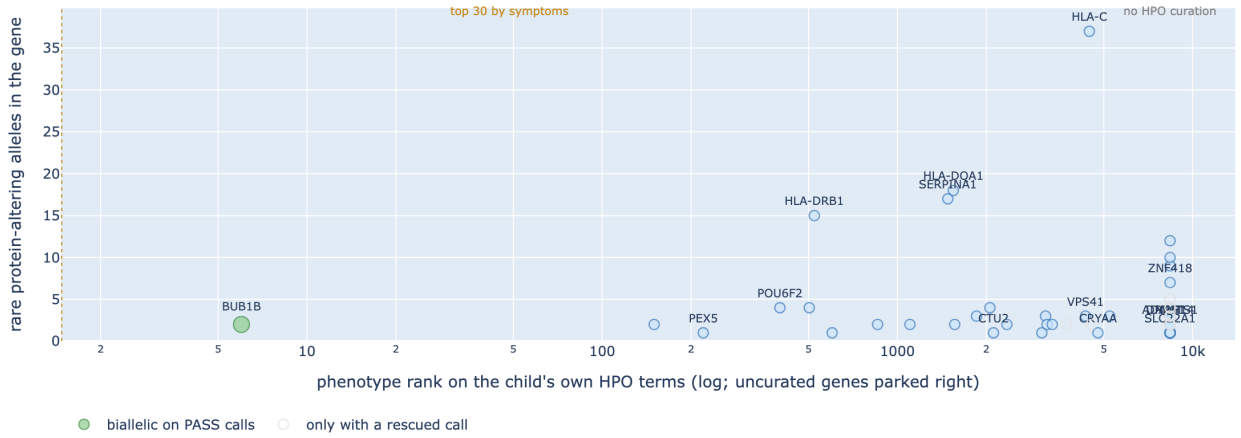


> **Legend** — **purple = the raw input** (every VCF record); **blue = machinery** (each filter); **green = the goal** (the last bar). Log axis: on a linear one every bar after the first would be invisible.

Stage	Survivors
variant records	5,012,204
distinct coding alleles, all annotated	38,087
rare	1,524
rare and protein-altering	875 (746 on PASS calls)
genes carrying any such allele	667
genes biallelic on PASS calls	**65**
… with a ClinVar pathogenic allele	**1**
… and phenotype rank ≤ 30	**1 — BUB1B**

The hypothesis was filed *before* the annotation ran (**H004**: BUB1B is the only gene satisfying all three conditions; other biallelic genes will exist but fail the second or third). It held: **supported 0.93**. The 64 other biallelic genes are what any genome shows — HLA genes with 11–18 "rare" alleles (hypervariable by construction), SERPINA1 with 16 (paralogous mapping), and two-missense carrier shapes with phenotype ranks in the hundreds or no curation at all (070).

5.3 Symptoms versus DNA (step 3, entry 071)



> **Legend** — **green = the goal** (the highlighted gene); **yellow = a decision point** (a gene carrying a ClinVar-pathogenic allele); **blue = machinery** (other biallelic genes); **grey = context** (genes biallelic only if a filtered call is rescued). Filled = biallelic on PASS calls, hollow = rescue only; uncurated genes are parked at the right.

Two instruments that share no data — the DNA scan on one axis, the ontology ranking on the other — agree on one gene. Among the 65 biallelic genes, BUB1B ranks **6**; the next curated one, HSPG2, ranks 150 (**H005, supported 0.80**). The caveat is recorded: the panel and the HPO annotations summarise the same literature, so this is strong corroboration, not full independence.

5.4 Nothing hides behind a filter (step 4, entries 049 and 072)

The caller suppressed 271k calls genome-wide, 1,594 of them inside coding intervals (`MQ40` 1,048, `QD2` 245, `LowQual` combinations 301). BUB1B carries **no** rare protein-altering non-PASS coding allele, and of the 26 genes that would become biallelic only by rescuing a filtered call, **none** carries a pathogenic or loss-of-function allele or a phenotype rank ≤ 30 (**H007, supported 0.80**; panel result **H006, retrospective, supported 0.77**). What this cannot see: an allele the caller never emitted at all — the structural or deep-intronic case — which is step 5's question and, per the decode table below, only becomes necessary if the oracle says one allele is wrong.

5.5 Spelled the way the scorer reads it (step 1, entry 073)

The scorer matches `(chrom, pos, ref, alt)` by exact string. Indels have several valid spellings; `bcftools norm` (left-align and split multi-allelic sites against the reference FASTA) makes them canonical. Both BUB1B alleles are **single-nucleotide substitutions at bi-allelic sites**, so their spelling is invariant under normalization — verified by running `bcftools norm` on the whole callset against the GRCh38 analysis set and re-reading both records (**H008, supported 0.83**, 073). Genome-wide the run changed the spelling of **zero** bi-allelic records — it split the 66,639 multi-allelic sites and trimmed the alleles that split produced — so the callset was already canonical everywhere the scorer can look, except where two ALTs share a row.

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6. The two alleles

Coordinates and the missense spelling are deliberately absent from this repository (Data Transfer Agreement); the stop_gained is a public ClinVar record (data-findings §6.4).

	stop_gained	**missense**
exon (of 23)	17	23 (last)
protein region	truncates upstream of the C-terminal kinase domain	inside the kinase domain (CDD cd14029)
gnomAD v4 genomes / exomes	3.29e-5 / 7.87e-5, no homozygote	**absent / absent**
ClinVar	**VCV000533901**, pathogenic / likely pathogenic, multiple submitters, no conflicts, *Mosaic variegated aneuploidy syndrome 1*	none
CADD phred	36	24.5
AlphaMissense	—	**0.92, likely pathogenic**
SpliceAI	0.03	0.00
genotype · depth · allele fraction	0/1 · 46× · 0.54	0/1 · 28× · 0.46
FILTER · QUAL	PASS · 709	PASS · 345

Mechanism. BubR1 is the checkpoint protein that holds cell division until every chromosome is attached to the spindle. Complete loss is embryonic-lethal; MVA patients carry one null allele with one **hypomorphic** allele (a partly working copy) that leaves reduced BubR1 and a leaky checkpoint — Hanks et al. 2004 (PMID 15475955), Suijkerbuijk et al. 2010 (PMID 20516114). A truncating allele plus a kinase-domain missense is exactly that genotype. Rhabdomyosarcoma is a reported MVA malignancy (PMID 31184400). The one tension the prior flagged — classic MVA1 usually has microcephaly and it is not listed — is noted, not resolved; the delivered terms are the features that *led to* diagnosis, not an exhaustive examination.

Phase. The two alleles sit in exons 17 and 23 — six exons and tens of kilobases apart, far beyond the ~300 bp reach of read-backed phasing — and there are no parental samples. Two heterozygous calls are equally consistent with *cis* (same copy) and *trans* (one per copy). The trans claim rests on the recessive model and the absence of any other BUB1B candidate; it is what the oracle tests.

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7. The submission, read as a measurement (step 6, entry 074)

The scorer returns `full_match_rank`, `partial_match_rank`, `f_max` and its threshold. Read against an explicit **hypothesis space** — every answer key still considered possible — a reply is a measurement, not a score. `mva.pipeline.probe_design` simulates the reply for every key with the local scorer replica, groups keys by the reply they produce (the **decode table**), and reports the expected information gain in bits.

Two facts the designer surfaced, both under test: a single allele ranked *below* the pair adds almost nothing (it cannot say which member of the pair it partners), while an alternative **pair** row at rank *r* decodes directly through `full_match_rank = r` and earns 50 or 25 points instead of 10. So ranks 2–5 hold pairs.

rank	epcr	type	row
1	0.95	primary	**BUB1B stop_gained + missense** (H009)
2–3	0.30 · 0.29	primary	stop / missense + the gnomAD-absent intronic BUB1B allele (SpliceAI 0.00)
4–5	0.20 · 0.19	primary	stop / missense + the rare intronic BUB1B allele (ClinVar VUS, SpliceAI 0.00)
6	0.05	secondary	**LZTR1** heterozygous stop_gained, ClinVar P/LP, gnomAD 1.4e-6 — dominant schwannomatosis predisposition, unrelated to MVA
7	0.04	secondary	**F5** Factor V Leiden, heterozygous — low-penetrance thrombosis risk
8	0.03	secondary	**FLG** heterozygous stop_gained, ClinVar P — ichthyosis vulgaris / atopy

full match rank	partial match rank	F-max	at epcr	prior mass	what survives
1	—	1.000	0.95	0.70	stop+missense
—	1	0.500	0.95	0.17	stop+unseen, missense+unseen
—	—	0.000	—	0.05	neither (other gene)
2	—	0.800	0.3	0.03	stop+intronA
3	—	0.800	0.3	0.02	missense+intronA
4	—	0.667	0.2	0.02	stop+intronB
5	—	0.667	0.2	0.01	missense+intronB

> **Legend — green = the pair confirmed at rank 1**; **yellow = a reply that needs a follow-up probe**; **red = nothing matched**; **blue = a decisive alternative**.

Rehearsed byte-for-byte on the replica **and re-graded with the organisers' published `evaluation.py`** — the two scorers agree on every one of the eight keys (075). If the pair is right, **100 points, F-max 1.0**. Seven distinguishable replies over eight keys; prior entropy 1.60 bits, **expected gain 1.46 bits**. The one residual ambiguity is structural — "one allele right, partner not in the callset" cannot say *which* allele — and it is the exact trigger for the raw-read route (step 5) and a two-row singles probe. Rows 6–8 are secondary findings: free under the scoring rules (rows below the hit cannot lower F-max), reviewed qualitatively, chosen by a stated rule (ClinVar P/LP after excluding conflicts and benign-tagged records, gnomAD < 0.02, dominant or risk-factor genes; recessive carrier states GNRHR, CIAO1, ZFYVE26 are reported here, not submitted).

Not uploaded. The four gate conditions of step 6 hold — normalized spellings, hypotheses with predictions and refutations, a byte-level rehearsal, four agreeing methods — and the timing is the team's decision. File: `build/submission/track1_submission_v1.csv` (gitignored; sha256 `fbc9e750…`).

8. Reproducibility, runtime and cost

```
uv sync --all-groups
uv run python tools/build_coding_regions.py --gtf Homo_sapiens.GRCh38.116.gtf.gz   # 5 s
uv run mva-warehouse stage && uv run mva-warehouse pg-ingest --tables coding_regions   # 23 s
uv run mva-warehouse pg-derive --only coding_alleles                                   # ~17 m
uv run mva-warehouse annotate-coding --workers 4                                       # ~33 m
uv run mva-warehouse pg-derive --only coding_candidates gene_biallelic                 # 24 s
uv run python tools/render_track1_figures.py
uv run python tools/resolve_submission_spec.py --selectors submissions/track1/iteration-1/spec
uv run python tools/design_submission.py --spec build/submission/spec.json --out build/submiss
```

The submission itself is reproducible without a single committed coordinate: every allele in `spec.selectors.json` is a *selector* (gene, consequence, ClinVar and gnomAD flags) that must match exactly one row of the warehouse; the resolver refuses anything else.

Everything ran on one laptop against a small hosted PostgreSQL and the free Ensembl REST API: **about one hour of wall clock, no paid compute**. The heavy alternative — aligning 160 Gb of reads — was not needed for the answer and stays conditional. `tools/check-python.sh` runs the eight quality gates (the last two are the coordinate gate and the iteration-folder gate); the new tooling adds 77 tests across eight new test modules, 1,687 in all.

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9. Limitations, stated

- **Coding-only scope.** The genome-wide scan covers CDS ± 8 bp (1.4 % of the genome). A deep-intronic, UTR or structural second allele in *another* gene is outside it. Within BUB1B the two only rare non-coding alleles were checked (SpliceAI 0.00).
- **Phase unresolved.** No reads or parents connect the two alleles; trans is inferred.
- **No mosaicism measurement.** Bulk blood WGS bounds the aneuploid fraction below ≈ 10–15 %; it does not measure it.
- **Missing feature.** Microcephaly, usual in MVA1, is not among the delivered terms.
- **Confidence numbers are a heuristic.** The register's weighing is transparent arithmetic for triage, not a calibrated posterior; evidence rows about the same two alleles are not independent, which is why H009 carries no committed confidence.
- **The oracle is the only authority.** The scorer replica is transcribed from the published code, tested against hand-derived values, and cross-checked key by key against the organisers' own `evaluation.py`; agreement between two scorers proves the *grading* is right, nothing about the answer key.

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10. Methods form (Track 1), answered

Question	Answer
Model	one approach, model 1: panel filter → genome-wide coding scan → phenotype join → pair submission
Automated or curated?	**Automated** to the ranked gene table and the candidate pair; the submission *layout* (which alternative pairings and secondary rows fill ranks 2–8) is a designed measurement, chosen by a stated rule and simulated, not hand-curated from belief
Public data only?	**Yes**: Ensembl 116 gene models and VEP (gnomAD v4, ClinVar, SpliceAI, AlphaMissense, CADD), HPO release 2026-06-23, PubMed. No proprietary data
Pairs or singles?	**Pairs**: every primary row is a compound-heterozygous pair in one row; the pipeline ranks pairs, not variants
Secondary findings	ClinVar P/LP after excluding conflicting and benign-tagged records, gnomAD AF < 0.02, dominant or risk-factor genes → three rows marked `secondary`; recessive carrier states reported, not submitted
Runtime and cost	~1 h wall clock on one laptop, free public APIs, one small hosted PostgreSQL; no paid compute
Abstract	§1 and §9 above, 300 words together

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11. Where every claim lives

Claim	Register	Test
Panel filter leaves exactly the BUB1B pair	H003, supported 0.93 (retrospective)	031, 051
No coding gene outranks the pair, genome-wide	H004, supported 0.93	069, 070
Symptoms and DNA agree on BUB1B	H005, supported 0.80	071
The parental term degrades the ranking	H002, supported 0.86	067
No filtered call hides a second allele	H006 (0.77, retrospective), H007 (0.80)	049, 072
Both alleles are spelled as the scorer reads them	H008	073
The causal pair is the BUB1B stop_gained + missense, in trans	H009, `testing`	074 → the oracle
The causal gene ranks ≤ 30 by symptoms	H001, `proposed`	settled by the oracle together with H009